Multi-Horizon Joint-Embedding Predictive Architectures and Coordinate Circuit-Breaker Panels for Multivariate Time-Series Anomaly Detection

Andre Williams, Imadeldin Mahgoub, *Senior Member, IEEE,* and Hamid Akbarian

***Abstract*—Reconstructive time-series anomaly detectors score raw observations, making them sensitive to benign jitter and drift. Joint-Embedding Predictive Architectures (JEPAs) forecast normalized latent summaries, yet collapsing coordinates into one scalar residual allows drifting coordinates to inflate thresholds. Targeting abrupt and intermediate-onset faults, we study two multi-horizon coordinate circuit-breaker architectures: Laminar-JEPA-Lite, our preferred deployment configuration, pairs a parallel three-horizon single-normalization predictor and channel-gated spectral splitting with 128 position-plus-drift switches and zero weight edits; and Laminar-JEPA, a four-horizon, 224-dial reference variant adding fluctuation, endpoint differences, 32 onset relays, and optional weight-space quenching. Laminar-JEPA-Lite cuts parameters by 35 to 52 percent and accelerates inference by 2.43 times. On the 33-stream development benchmark, Laminar-JEPA and Laminar-JEPA-Lite hold autonomous causal false-positive rate (FPR) at 1.3 and 1.0 percent versus 22.2 to 24.3 percent for reconstructors, reaching buffered point-wise F1 of 0.244 and 0.242 and seven-dataset macro F1 of 0.133 and 0.171. On the 19-stream cache, 224 dials lift causal F1 from 0.013 to 0.071 over scalar collapse, and 160 dials reach 0.098. Across 118 untouched held-out streams under a five-block protocol separating checkpoint selection and nominal adaptation from threshold calibration, Laminar-JEPA-Lite lifts causal F1 from 0.076 to 0.079, buffered F1 from 0.143 to 0.156, and causal event recall from 52.5 to 63.8 percent across all nine datasets, outperforming system-matched coordinate maximum and top-k banks. Calibration precision depends on autocorrelation-adjusted sample size; causal ramp detection vanishes at four context windows, and single-sensor sensitivity remains an optional trade-off.**

***Index Terms*—Anomaly detection, Multivariate time series, Representation learning, Self-supervised learning, Statistical process control.**

I. Introduction

Telemetry monitors track dozens of channels simultaneously [1]–[4]. Because labeled faults are scarce, a streaming detector must learn nominal behavior from healthy history alone, flag anomalies promptly, and keep false alarms low during normal operation.

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Why Observation-Space Reconstructors Suffer High False-Alarm Rates: Reconstructors such as TimesNet [5], TranAD [6], and Anomaly Transformer [7] score raw sensor error. Because raw telemetry mixes jitter, thermal drift, and disparate channel scales, summing errors across 25–55 channels produces heavy-tailed spikes. Calibrated on healthy windows via Streaming Peaks-Over-Threshold (SPOT) [8], reconstructors fire during 22%–31% of later healthy data; a rate-matched coin flip catches 60.0% of anomaly events, exceeding TimesNet's 58.8%.

The Bottleneck of Scalar Score Aggregation: Joint-Embedding Predictive Architectures (JEPAs) [9], [10] compress each window into a 32-dimensional normalized summary and forecast future summaries in latent space, cutting nominal false alarms to 0.6%–2.4%. Yet scalar latent detectors still sum all coordinate errors against one threshold: if one or two coordinates shift slightly between training and calibration (*hijacker dials*), they inflate the global threshold and mask faults on the other 30 coordinates (causal Point-F1 0.050–0.055, delay 138–154 steps).

*Multi-Horizon Coordinate Circuit Breakers (*Laminar-JEPA-Lite *and* Laminar-JEPA*):* Targeting abrupt and intermediate-onset telemetry faults, we study a two-tier family of multi-horizon JEPA circuit-breaker architectures (Fig. 1): (1) *Laminar-JEPA-Lite*, our preferred deployment configuration distilled from a layer-by-layer activation and component autopsy, pairs a parallel three-horizon single-layer-normalization (LayerNorm) [11] predictor and channel-count-gated spectral splitting ($K \geq 25$) with $4 \times 32 = 128$ position-plus-drift breakers under one nominal quorum and zero weight editing, cutting parameters by 35%–52% and accelerating windowed inference by $2.43\times$; and (2) *Laminar-JEPA*, retained as the expressive four-horizon, 224-dial reference architecture, expands each coordinate's error trajectory into $7 \times 32 = 224$ dials plus 32 right-edge onset relays with optional split-gated weight-space quenching.

A. Contributions

We separate evaluation into a 33-stream development cohort (plus a 19-stream residual cache) and a disjoint 118-stream untouched held-out cohort within a 141-stream archive across 12 datasets under a strictly disjoint five-block calibration protocol (Section V-B):

1) *Preferred distilled and reference panels:* On the 33-stream benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach

buffered Point-F1 of 0.244 and 0.242 and causal event recall (ER) of 67.3% and 65.0% at 1.3% and 1.0% held-out false-positive rate (FPR), with *Laminar-JEPA-Lite* leading 7-dataset macro Point-F1 (0.171 vs. 0.133). On the 19-stream cache, the 224-dial panel lifts causal Point-F1 from 0.013 to 0.071 over 1D collapse ($p = 0.0023$), and a 160-dial subset reaches 0.098. Across 118 untouched streams ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.076 to 0.079 ($p = 0.014$; $+0.022$ over TS-JEPA, $p = 1.8 \times 10^{-4}$), buffered Point-F1 from 0.143 to 0.156 ($p = 0.022$), and causal ER from 52.5% to 63.8% across all $9/9$ datasets (sign-flip $p = 0.0039$; $12/12$ leave-one-dataset-out (LODO) folds on ER and buffered F1), while outperforming system-matched coordinate $\max_j z_j$ (0.062, $p = 3 \times 10^{-5}$) and unguarded top-k voting (0.073, $p = 0.0034$).

2) *Autopsy distillation, mechanisms, and latency:* Matched-backbone controls (Table II(c)) isolate LayerNorm tail compression and barycenter drift reduction while showing scalar spectral reweighting is null. Layer activation traces, LayerNorm spillover audits, and a $K \geq 25$ spectral threshold motivate *Laminar-JEPA-Lite*. Across $S \in \{1, 4, 16, 32, 64\}$ on five exploratory streams, both panels reach 2.1–$4.3\times$ the scalar causal Point-F1 at $S = 1, 4$ without establishing cross-domain latency rules.
3) *False-alarm control and contextual faults:* On the 19 non-GHL streams, latent prediction holds autonomous causal FPR at 0.6%–2.4% versus 22.2%–24.3% for reconstructors ($N = 7$ dataset $p = 0.016$). At diagnostic budget $f^* = 1\%$ test FPR, *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% and 56.0% of events versus 21.7% for TimesNet, and reach 0.519–0.532 buffered precision-recall (PR) AUC with 77.5%–87.5% autonomous ER on NASA contextual faults (Tables IV–V, Fig. 4).
4) *Calibration sample size, single-channel branch, and limits:* Proposition 1 bounds horizon-fluctuation concentration $R_{\text{LT}} \in [\frac{1}{M-1}, 1]$, while scalar R_{LT} reweighting is null ($\Delta\text{F1} = -0.005$). Sokal autocorrelation (τ_{int} up to 206) and block-bootstrap audits show $N_{\text{eff}} = 300$ does not guarantee 99.9th-percentile precision; interquartile-range (IQR) and Beta-smoothed quantiles remain exploratory. Across 19 streams, an optional channel branch lifts injected single-sensor ER to 80.7%–96.5% while raising nominal causal FPR from 1.5% to 6.8%–19.4%; natural sensor failures and slow ramps ($L \geq 4C$) remain limitations.

II. Related Work

Observation-Space Forecasting and Reconstructive Detectors: Deep detectors typically forecast or reconstruct raw windows and threshold $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$: autoregressive long short-term memory (LSTM) forecasters [4] model spacecraft channels directly; TranAD [6] pairs adversarial training with self-conditioning attention; Anomaly Transformer [7] contrasts Gaussian prior and series-level attention; and TimesNet [5] reshapes 1D series along Fast Fourier Transform (FFT) periods for 2D convolution. Equal channel-wise reconstruction losses fit stochastic jitter, so summing errors across $K \geq 25$ channels yields heavy-tailed (*leptokurtic*, $\gamma_2 > 0$) spikes and high false alarms under baseline shift [12].

Contrastive and Joint-Embedding Predictive Architectures: Representation methods encode windows into compact latents [13]. Contrastive approaches—temporal-contextual contrasting [14], TS2Vec [15], Neural Contextual Anomaly Detection (NCAD) [10] with synthetic anomalies, and dual-attention DCdetector [16]—depend on whether synthetic perturbations or patch discrepancies match test faults (0.088 and 0.097 stream Point-F1 for NCAD and DCdetector vs. 0.186 for latent prediction on 130 streams, Section VII-A). Joint-Embedding Predictive Architectures (JEPAs) [9], [17], [18] predict target embeddings z_{tgt} from context z_{ctx} without observation decoding or negative pairs, avoiding collapse via exponential moving average (EMA) targets [19] or variance-covariance terms [20]–[22].

Residual Decomposition, Conformal Thresholding, and Statistical Process Control: Telemetry reflects underlying system dynamics [23]. Neural ordinary differential equations (ODEs) [24], Physics-Informed Neural Networks (PINNs) [25], Hamiltonian/Lagrangian nets [26], [27], and Koopman lifts [28]–[32] model structured dynamics, though partial differential equation (PDE) penalties require known laws and exhibit stiff gradients on uncalibrated sensors [33]; we instead borrow the algebraic mean-plus-fluctuation identity of Reynolds averaging [34] strictly as a multi-horizon residual summary. Multivariate statistical process control (SPC) charts such as Hotelling's T^2 [35] and Multivariate Exponentially Weighted Moving Average (MEWMA) [36] collapse whitened coordinates into one scalar, as do 1D deep scores calibrated via extreme-value theory [37] and Generalized Pareto Distribution (GPD) tails in SPOT [8] under serial dependence [38]. Conversely, univariate Cumulative Sum (CUSUM) banks [39] and per-coordinate split-conformal rules under Bonferroni, Šidák, or Benjamini–Hochberg (BH) false-discovery-rate (FDR) control [40] hit a discrete calibration floor $(N_{\text{cal}} + 1)^{-1}$ on short histories, whereas matched-rate per-coordinate and recalibrated Mahalanobis conformal rules inflate under coordinate drift (Section VI-A). All methods are evaluated without point adjustment [12], [41], [42].

III. Multi-Sensor Telemetry Benchmark Suite

Our benchmark suite spans 45 continuous telemetry streams across 11 datasets from NASA spacecraft archives [4], the University of California, Irvine (UCI) Machine Learning Repository [43], [44], TimeEval [12], and mTSBench [45], expanded in archive-wide evaluations to 141 qualifying multivariate streams across 12 datasets (Section V-B). Thirty-three streams enter the development protocol (19 non-GHL streams and 14 GHL streams) and 12 are reserved with explicit accounting below. *Spacecraft and solar telemetry* covers six SMAP streams (K=25 channels), six MSL rover streams (K=55, four primary and two short-nominal), and SWAN-SF solar magnetograms [46] (K=38), where NASA [4] labels each SMAP/MSL sequence as an out-of-range *point* fault or in-range *contextual* fault (Table V(a)). *Wearable kinematics* covers six Daphnet Freezing-of-Gait (FoG) streams [43] at

64 Hz (K=9, 72 episodes, where S=64 equals 1 s). *Process and infrastructure telemetry* covers 14 Gasoil Heating Loop (GHL) streams (K=16, 26 slow-ramp faults), the Genesis pick-and-place testbed (K=18), GECCO water chemistry (K=9, 37 anomalies), Metro I-94 traffic (K=5), Room-Occupancy sensing [47] (K=5), CalIt2 building flow [44] (K=2), and six CICIDS network streams (K=72). Section VI-B evaluates six injected single-channel fault classes across all 19 multivariate development streams ($K \in [5, 55]$).

Each stream provides nominal history ($T_{\text{nom}} \in [1{,}920, 132{,}481]$ across the archive; $[3{,}769, 120{,}000]$ on the 33-stream cohort) and labeled test data ($N_{\text{test}} \in [2{,}158, 114{,}001]$), standardized per channel on the weight-fitting prefix ($\boldsymbol{\sigma}$ floored at 10^{-2}) and split chronologically into a 65% model-fitting prefix (subdivided on archive benchmarks into isolated weight-fitting $\mathbf{X}_{\text{fit}}$, checkpoint-selection $\mathbf{X}_{\text{sel}}$, and nominal adaptation $\mathbf{X}_q$ blocks; Section V-A), nominal threshold calibration $\mathbf{X}_{\text{cal}}$ (17.5%), a 64-step isolation gap, and held-out nominal evaluation $\mathbf{X}_{\text{hold}}$ (17.5%). Development 33-stream inclusion required $N_{\text{cal}} \geq 500$ ($T_{\text{nom}} \geq 3{,}769$), $K \leq 55$, and $N_{\text{test}} \leq 1.2{\times}10^5$ (Table IV(c)); the 12 reserved streams comprise five short-nominal streams ($T_{\text{nom}} \in [764, 1{,}492]$, $N_{\text{cal}} \leq 101$: CalIt2, MSL C-2, MSL D-16, two Room-Occupancy), six CICIDS streams (K=72), and GECCO (K=9, included in archive cohorts). Archive cohorts evaluate all 141 qualifying streams across 12 datasets ($N_{\text{cal}} \geq 100$, ≥ 1 anomaly: 47 SMAP, 22 Daphnet, 18 MSL, 18 SMD, 17 Exathlon, 13 OPPORTUNITY, GECCO, Genesis, Pooled Server Metrics (PSM), CreditCard, Metro, and SWAN; 118 untouched by design).

Two caveats bound external validity: SMAP and MSL contain documented labeling flaws [42], and streams within multi-stream archives share hardware platforms. Dataset clusters ($N \in \{7, 9, 12\}$) are therefore treated as co-primary units alongside streams via Dataset-Macro means, cluster-bootstrap confidence intervals (CI), exact dataset sign-flip permutation tests, and LODO audits (Section V-A).

IV. Methodology: Multi-Horizon Laminar-JEPA and Distilled Laminar-JEPA-Lite

A. Stage 1: Multiscale Windowing, Multi-Depth TCN Hierarchy Tapping, and Barycenter Centering

Fig. 1 illustrates the four-stage pipeline from raw telemetry $\mathbf{X} \in \mathbb{R}^{T \times K}$ (K channels over T timesteps) to binary alarm decisions. A sliding window of length $W = C + S = 320$ advances one step at a time, splitting each position t into a historical *context window* $\mathbf{x}_{\text{ctx}}$ of $C = 256$ steps and a future *target horizon* $\mathbf{x}_{\text{tgt}}$ of $S = 64$ steps (ending at the causal decision timestamp $\tau = t + S - 1$). Before mixing channels, multichannel inputs ($K > 1$) pass through a causal depthwise 1D convolution ($K \leq 1$ uses an identity bypass) and are split across four smooth dyadic frequency shells on a 256-point real fast Fourier transform (rFFT) grid. The shells form an amplitude partition of unity within $1.2{\times}10^{-7}$, separating slow baseline drift from high-frequency vibration before per-shell causal 1D convolutions (kernels $\{15, 9, 5, 3\}$) and a softmax router feed a causal dilated temporal convolutional network (TCN) [48].

Rather than reconstructing raw sensor waveforms, a Joint-Embedding Predictive Architecture (JEPA) [9] maps context and target windows into compact $D = 32$-dimensional layer-normalized summaries via dual online (E_θ) and target (E_ϕ) encoders. Target weights track online weights via an exponential moving average (EMA) at momentum $m = 0.996$ (0.995 on archive cohorts) [17], [19], placing summaries near radius $\sqrt{D}$ up to the normalization epsilon. Deep dilated TCN stacks can smooth out brief onset spikes, while operating-regime shifts displace the latent mean. Both encoders address these two failure modes directly:

1) *Multi-depth hierarchy tapping:* Each encoder taps three intermediate feature maps ($d_{\text{model}} = 48$) from the shallow spectral stem, mid-depth TCN block, and final TCN block. Concatenating the channel-centered terminal state and temporal root-mean-square (RMS) energy of each tap, projecting to $\mathbb{R}^D$ through a two-layer GELU head, and blending with the deep TCN output through a bounded gate ($g \leq 0.5$) preserves fast transients alongside long-range context.
2) *Synchronized barycenter centering:* During training, an online centroid $\bar{\boldsymbol{\mu}}_{\text{ctx}} \in \mathbb{R}^D$ updates at rate 0.05, copies synchronously to E_ϕ, and is subtracted from both context and target summaries prior to final LayerNorm. Centering reduces learned common-mode offsets without guaranteeing invariance to unseen drift.

B. Stage 2: Nested Causal Multi-Horizon Rollout and Variance-Floor Decorrelation Objective

In the single-horizon baseline (*TS-JEPA*), a two-layer multilayer perceptron (MLP) predictor maps z_{ctx} to a single 64-step forecast $\hat{z}_{\text{tgt}}$, trained with predictive mean-squared error (MSE) and a VICReg-style log-determinant volume penalty ($\beta = 0.05$) [20]. Pooling an entire 64-step target window into one vector can average out short-lived faults, and LayerNorm bounds vector length without bounding temporal tail heaviness.

To track how prediction error evolves across the target window, Stage 2 of *Laminar-JEPA* encodes $M = 4$ nested target prefixes of lengths $S_m \in \{16, 32, 48, 64\}$ at $S = 64$ (clamped to at least two steps when $S > 1$; at $S = 1$, all four prefixes share the single target step and differ only through predictor rollout). Starting from z_{ctx} with learned horizon embeddings $\mathbf{p}_m \in \mathbb{R}^D$, a recurrent sub-horizon predictor P_{sub} rolls out sequential LayerNorm forecasts $\hat{z}_{\text{tgt}}^{(1)}, \ldots, \hat{z}_{\text{tgt}}^{(M-1)}$ while an anchor head P_θ predicts the full horizon $\hat{z}_{\text{tgt}}^{(M)}$. Assigning half the loss weight to $m = M$ and splitting the other half equally across $m < M$, training minimizes multi-horizon MSE plus a non-collapse penalty $\boldsymbol{\Delta}_\gamma$:

$$\mathcal{L}_{\text{Lam}} = \sum_{m=1}^{M} \frac{w_m}{D} \Big(\big\| \hat{z}_{\text{tgt}}^{(m)} - \text{sg}\big(z_{\text{tgt}}^{(m)}\big) \big\|_2^2 + \big\| \boldsymbol{\Delta}_\gamma\big(\hat{z}_{\text{tgt}}^{(m)}\big) \big\|_F^2 \Big), \tag{1}$$

where $\text{sg}(\cdot)$ stops gradients through E_ϕ. Unlike log-determinant penalties that require batch size $B > D$, $\boldsymbol{\Delta}_\gamma(\mathbf{Z}) \in \mathbb{R}^{D \times D}$ penalizes collapse directly on small batches: its diagonal entries $\sqrt{\max(0, \gamma - \sigma_d)}$ enforce a per-coordinate

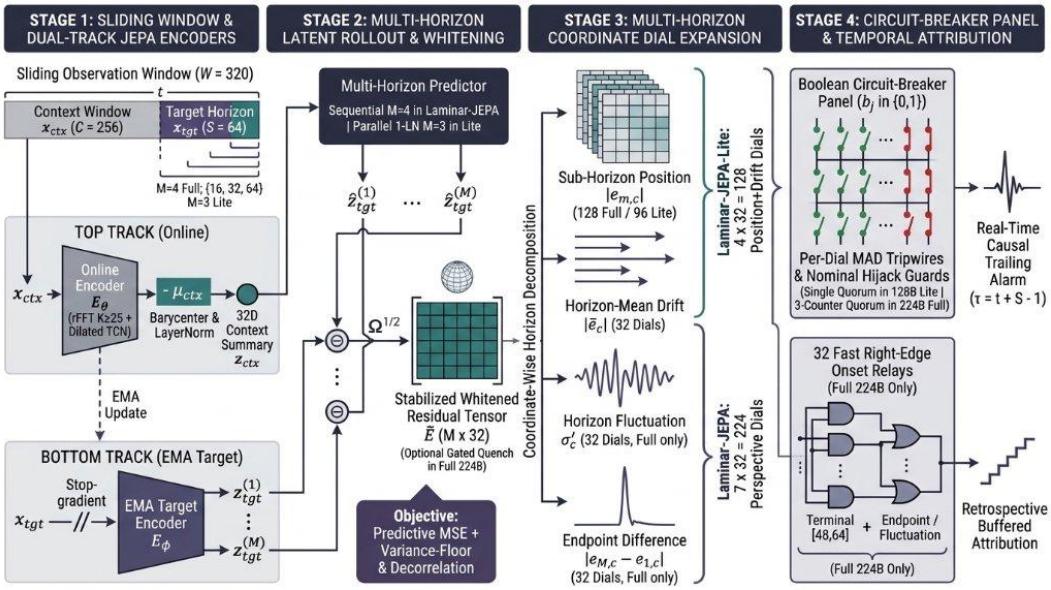


Fig. 1. Overview of the four-stage *Laminar-JEPA* and distilled *Laminar-JEPA-Lite* pipeline: **(1)** sliding window (C=256, $S \in \{1..64\}$) and dual encoders (E_θ, E_ϕ) with conditional rFFT splitting ($K \geq 25$ in Lite), barycenter centering, and LayerNorm; **(2)** multi-horizon rollout (M=4 sequential in full *Laminar-JEPA*; M=3 parallel single-LayerNorm in *Lite*) under variance-floor and decorrelation loss plus resolvent whitening into $\tilde{\mathbf{E}} \in \mathbb{R}^{M\times 32}$; **(3)** horizon expansion into $7 \times 32 = 224$ dials (full) or $4 \times 32 = 128$ position-plus-drift dials (*Lite*); and **(4)** per-dial Boolean circuit breakers emitting causal (τ=t+S−1) or prefix-buffered alarms.

standard-deviation floor $\gamma = 0.75$ (using batch-biased variance plus 10^{-4}), and its off-diagonal entries $\Sigma_{dj}/\sqrt{2}$ penalize cross-coordinate sample covariance ($B-1$ divisor). At $B = 16 < D = 32$, effective latent rank $\mathrm{RankMe}(z_{\mathrm{tgt}})$ rises from 11.6 to 19.8 [49].

C. Stage 3: Calibration-Stabilized Whitening, Optional Weight-Space Quenching, and 224-Dial Expansion

Collapsing a multi-horizon residual tensor into a single scalar score creates a bottleneck: if nominal drift inflates even a few coordinates between training and calibration, those coordinates dominate the scalar norm. Stacking the $M = 4$ signed residuals $\mathbf{e}_m(t) = z_{\mathrm{tgt}}^{(m)}(t) - \hat{z}_{\mathrm{tgt}}^{(m)}(t)$ forms $\mathbf{E}(t) \in \mathbb{R}^{M\times D}$ ($MD = 128$ entries). On nominal training windows, eigendecomposing the residual covariance $\hat{\mathbf{\Sigma}}_{\mathrm{res}} = \mathbf{U}\,\mathrm{diag}(\lambda_k)\mathbf{U}^\top$ defines a damped resolvent precision matrix $\mathbf{\Omega} = \mathbf{U}\,\mathrm{diag}\big(\lambda_k/(\lambda_k + 10^{-3})^2\big)\mathbf{U}^\top$. Multiplying centered residuals by $\mathbf{\Omega}^{1/2}$ yields whitened errors $e_{m,c}^{\mathrm{white}}(t)$ while suppressing near-null eigenmodes instead of amplifying them. Stage 3 then insulates coordinates against nominal calibration drift and expands the horizon trajectory:

1) *Disjoint nominal stabilization and optional quenching on* $\mathbf{X}_q$*:* Estimating blending weights or weight edits on the same block $\mathbf{X}_{\mathrm{cal}}$ that sets final alarm thresholds would couple model adaptation to threshold calibration. Under our disjoint adaptation protocol ($\mathbf{X}_q \perp \mathbf{X}_{\mathrm{cal}}$, Section V-A), a separate nominal block $\mathbf{X}_q$ fits raw residual RMS scales and smoothly blends any coordinate whose whitened RMS exceeds 1.10 toward its standardized raw residual $\hat{e}_{m,c}^{\mathrm{raw}}(t)$, freezing all weights before $\mathbf{X}_{\mathrm{cal}}$ is touched:

$$\tilde{e}_{m,c}(t) = w_{m,c}\, e_{m,c}^{\mathrm{white}}(t) + (1 - w_{m,c})\, \hat{e}_{m,c}^{\mathrm{raw}}(t),$$
$$w_{m,c} = \frac{1}{1 + 2\max\big(0, \mathrm{RMS}_q^{\mathrm{white}}(m,c) - 1.10\big)^2}. \quad (2)$$

Full *Laminar-JEPA* also runs an optional zero-gradient *weight-space quench* strictly on $(\mathbf{X}_{\mathrm{fit}}, \mathbf{X}_q)$ prior to $\mathbf{X}_{\mathrm{cal}}$ (legacy fixed-epoch development runs without $\mathbf{X}_q$ estimate both on $\mathbf{X}_{\mathrm{cal}}$, Table I). Coordinates are ranked by a median/median-absolute-deviation (MAD) standardized instability score combining $\mathbf{X}_{\mathrm{fit}} \to \mathbf{X}_q$ mean/scale shifts, $\mathbf{X}_q$ half-split drift, residual autocorrelation, and a negative guard for isolated spikes. For candidate top-k drifting coordinates ($k \in \{4, 2\}$), four steps run sequentially: *(Q1)* damping selected predictor output rows/LayerNorm gains and shifting $\mathbf{p}_m$ toward $\mathbf{X}_q$ means (inducing unselected spillover through 32D LayerNorm, audited in Section VI-A); *(Q2)* blending residual centers 75% toward post-edit $\mathbf{X}_q$ means; *(Q3)* attenuating off-diagonal precision couplings by 0.15 and rescaling whitened $\mathbf{X}_q$ RMS; and *(Q4)* retaining the edit (preferring $k = 2$ on ties) iff $\mathbf{X}_q$ half-split FPR improves while preserving $\geq 92\%$ of pre-edit quorum thresholds, freezing all parameters thereafter.

2) *Multi-horizon expansion into* $7D = 224$ *dials:* Inspired algebraically by the mean-plus-fluctuation split of Reynolds averaging [34] across the $M = 4$ horizons of each coordinate $c \in \{1, \ldots, 32\}$, we separate $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{4\times 32}$ into its horizon mean $\bar{e}_c(t) = M^{-1}\sum_{m=1}^{M} \tilde{e}_{m,c}(t)$ and zero-mean fluctuation $F_{m,c}(t) = \tilde{e}_{m,c}(t) - \bar{e}_c(t)$. Taking magnitudes yields $7D = 224$ nonnegative dials $\mathbf{d}(t) \in \mathbb{R}^{224}_{\geq 0}$: (a) $4D = 128$ *sub-horizon positions* $|\tilde{e}_{m,c}(t)|$; (b) $D = 32$ *horizon-mean drifts* $|\bar{e}_c(t)|$; (c) $D = 32$ *horizon fluctuations* $d_{\mathrm{fluc},c}(t) = \sqrt{M^{-1}\sum_{m=1}^{M} F_{m,c}^2(t)}$; and (d) $D = 32$ *endpoint differences* $d_{\mathrm{accel},c}(t) = |\tilde{e}_{M,c}(t) - \tilde{e}_{1,c}(t)|$.

Removing the horizon mean splits each coordinate's squared error orthogonally into persistent offset plus within-window fluctuation, and confines the $M \times M$ horizon Gram matrix to the $(M-1)$-dimensional subspace orthogonal to $\mathbf{1}_M$ (a geometric property of the representation rather than an accuracy guarantee):

Proposition 1 (Horizon Energy Split and Spectral Concentration Bounds)**.** *For $\tilde{\mathbf{E}}(t) \in \mathbb{R}^{M\times D}$ ($M \geq 2$) with horizon mean $\bar{\mathbf{e}}(t)$ and fluctuation matrix $\mathbf{F}(t) = \tilde{\mathbf{E}}(t) - \mathbf{1}_M \bar{\mathbf{e}}(t)^\top \neq \mathbf{0}$: (i) for every coordinate c, $M^{-1}\sum_{m=1}^{M} |\tilde{e}_{m,c}(t)|^2 = |\bar{e}_c(t)|^2 + d_{\mathit{fluc},c}^2(t)$ and $d_{\mathit{accel},c}(t) \leq \sqrt{2M}\, d_{\mathit{fluc},c}(t)$; and (ii) the $M \times M$ horizon Gram matrix $\mathbf{G}(t) = \mathbf{F}(t)\mathbf{F}(t)^\top \in \mathcal{S}_+^M$ has ordered eigenvalues $0 = \lambda_1 \leq \lambda_2 \leq \cdots \leq \lambda_M$ whose scale-invariant dominant-mode share for any $p > 0$ [50] satisfies*

$$R_{LT}\big(\mathbf{F}(t)\big) = \frac{\lambda_M^p}{\sum_{m=1}^{M} \lambda_m^p} \in \Big[\frac{1}{M-1},\, 1\Big], \tag{3}$$

reaching $\frac{1}{M-1}$ iff energy is isotropic across all $M-1$ active modes ($\lambda_2 = \cdots = \lambda_M > 0$, requiring $D \geq M-1$) and 1 iff $\mathbf{F}(t)$ has rank one. Empirically, scalar reweighting by R_{LT} has a null effect (Section VI).

Proof. Zero column sums ($\sum_{m=1}^{M} F_{m,c} = 0$) cancel the cross term in $M^{-1}\sum_{m=1}^{M}(\bar{e}_c + F_{m,c})^2$, and $(F_{M,c} - F_{1,c})^2 \leq 2(F_{M,c}^2 + F_{1,c}^2) \leq 2M\, d_{\mathit{fluc},c}^2$ gives (i). For (ii), $\mathbf{F}^\top \mathbf{1}_M = \mathbf{0}_D$ implies $\mathbf{G}\mathbf{1}_M = \mathbf{0}_M$ ($\lambda_1 = 0$), while $\|\mathbf{F}\|_F^2 > 0$ implies $\lambda_M > 0$. Summing $0 \leq \lambda_m^p \leq \lambda_M^p$ over $m \in \{2, \ldots, M\}$ yields $\lambda_M^p \leq \sum_{m=2}^{M} \lambda_m^p \leq (M-1)\lambda_M^p$; inverting and multiplying by λ_M^p gives (3) and its equality conditions, and nonzero scalar rescaling cancels. □

D. Stage 4: Zero-Collapse 224-Bit Circuit-Breaker Panel, Onset Relays, and Temporal Attribution

Instead of adding all 224 dials into one global score, Stage 4 (Fig. 1, Algorithm 1) assigns each dial its own binary circuit breaker on nominal calibration data. Because the 224 dials are derived from a single 4×32 residual tensor, their trips are correlated (median effective rank 65 of 224 on the development cohort, with joint trip-count variance $7.2\times$ independent trials). Alarm quorums are therefore calibrated directly on the empirical distribution of simultaneous nominal trip counts:

1) *Per-dial standardization and hijack guards:* Each dial $j \in \{1, \ldots, 224\}$ is standardized against its nominal calibration median m_j and scaled MAD $s_j = 1.4826\ \mathrm{MAD}(d_{j,\mathrm{cal}}) + 10^{-4}$ into a nonnegative score $z_j(t) = \max\big(0, (d_j(t) - m_j)/s_j\big)$. To quarantine coordinates that drift during calibration, its 98.5th-percentile tripwire τ_j is multiplied by two nominal hijack guards ($\gamma_j^{\mathrm{cal}} \geq 1$ when second-half calibration 98th percentiles exceed the first half by $> 1.35\times$, and $\gamma_j^{\mathrm{sc}} \geq 1$ when dial scale exceeds $1.5\times$ its perspective-group median by $> 1.25\times$):
$$b_j(t) = \mathbb{I}\big(z_j(t) > \tau_j\big), \quad \tau_j = q_{0.985}(z_{j,\mathrm{cal}})\, \gamma_j^{\mathrm{cal}}\, \gamma_j^{\mathrm{sc}}. \tag{4}$$
2) *32 right-edge onset relays:* When an abrupt fault first enters the right edge of the S-step target window, earlier prefixes ($m \in \{1, 2\}$) still cover nominal data. To catch these transitions promptly, each coordinate $c \in \{1, \ldots, 32\}$ defines stricter 99.3rd-percentile edge switches ($\gamma^{1.25}$ guard) and fires a binary onset relay $r_c(t) \in \{0, 1\}$ whenever a terminal-prefix trip ($m = 4$) coincides with an acceleration, fluctuation, or $m = 3$ trip on the same coordinate.
3) *Causal vs. buffered attribution, score weighting, and quorum voting:* Under *causal trailing attribution*, active breakers on window $[t, t+S)$ emit a real-time decision strictly at its right endpoint $\tau = t+S-1$ (structural lag $S-1$, empirical delay $\Delta t = t_{\mathrm{alm}} - t^*$). Under *retrospective buffered attribution*, position votes route to their corresponding prefixes $[t, t+S_m)$, drift and fluctuation votes span $[t, t+S)$, and endpoint-difference votes route to $[t+S_3, t+S)$ (empty at $S = 1$). Summing active trips yields $N_{224}(\tau)$, position subpanel count $N_{128}(\tau) \subset N_{224}(\tau)$, and relay count $N_{\mathrm{edge}}(\tau)$. Because N_{128} is contained in N_{224}, the continuous ranking score $N_{\mathrm{tot}}(\tau) = N_{224}(\tau) + N_{128}(\tau) + 2N_{\mathrm{edge}}(\tau)$ gives sub-horizon position trips and onset relays $2\times$ weight (fixed on development streams before held-out testing; unweighted N_{224} shifts PR-AUC by ≤ 0.003, Section VI-A), whereas binary alarms fire whenever any counter exceeds its nominal quorum:
$$\begin{aligned}\hat{y}(\tau) = \mathbb{I}\big(&N_{224}(\tau) > K_{224} \ \vee\ N_{128}(\tau) > K_{128} \\ &\vee\ N_{\mathrm{edge}}(\tau) > K_{\mathrm{edge}}\big),\end{aligned} \tag{5}$$
with nominal calibration quorums $K_{224} = \max(1.12\, q_{0.995}, 5.0)$, $K_{128} = \max(1.12\, q_{0.995}, 3.5)$, and $K_{\mathrm{edge}} = \max(1.25\, q_{0.999}, 2.5)$.

E. Autopsy-Distilled Architecture: Laminar-JEPA-Lite

Layer-by-layer activation tracing and component ablations of *Laminar-JEPA* (Section VI) motivate three simplifications that cut trainable parameters by 35%–52% (141.9k at $K < 25$; 222.3k at $K \geq 25$ vs. 295.8k) and accelerate windowed GPU inference by $2.43\times$ ($3.14\times$ at $K < 25$) with zero post-training weight editing:

1) *Four-block TCN and $K \geq 25$ spectral gating:* Because a fifth dilated block damps short-prefix anomaly divergence without improving effective rank, *Laminar-JEPA-Lite* uses four TCN blocks ($d \in \{1, 2, 4, 8\}$), 64D pooling heads, and activates the four-shell rFFT front-end only on high-channel streams ($K \geq 25$), using a direct causal projection when $K < 25$.
2) *Parallel three-horizon single-LayerNorm predictor:* Instead of sequential four-horizon unrolling and stacked predictor normalizations, a single residual two-layer GELU MLP predicts $M = 3$ nested prefixes ($S_m \in \{16, 32, 64\}$ at $S = 64$; 96×96 precision $\mathbf{\Omega}$) in parallel from $z_{\mathrm{ctx}} + \mathbf{p}_m$ with one terminal LayerNorm, avoiding sequential lag and LayerNorm cross-coupling.
3) *128-dial position-plus-drift panel and single quorum:* Retaining only the 96 sub-horizon magnitudes $|\tilde{e}_{m,c}(t)|$ and 32 horizon-mean drift dials $|\bar{e}_c(t)|$ ($4D = 128$ breakers calibrated via (4)) removes fluctuation/endpoint dials, onset relays, and weight quenching, ranking steps by unweighted $N_{128}(\tau)$ and firing under one quorum $K_{\mathrm{Lite}} = \max\big(1.12\, q_{0.995}(N_{128,\mathrm{cal}}),\, 3.5\big)$.

1D Scalar Collapse Controls: To isolate circuit-breaker gains over scalar aggregation on identical backbones (Section VI), we evaluate 1D controls that pool a residual vector via an equal blend of its ℓ_2 norm and softmax-weighted coordinate RMS ($\alpha = \tau_g = 0.5$). Single-horizon TS-JEPA applies this scalar pooler to $\mathbf{e}(t) \in \mathbb{R}^{32}$, whereas 1D Multi-Horizon JEPA combines the full-tensor and peak-horizon pooled scores in root-mean-square (with optional R_{LT} modulation $\alpha_{\mathrm{LT}} \in \{0.12, 0\}$), calibrated via SPOT [8], [37], [51] ($u = p_{98}, q = 10^{-3}$).

Algorithm 1 Laminar-JEPA (224-Bit) and Laminar-JEPA-Lite (128-Bit)

Require: Nominal history $\mathbf{X}_{\text{nom}}$, test stream $\mathbf{X}_{\text{test}}$, tier $\in \{\text{Full}, \text{Lite}\}$, attribution mode $\in \{\text{causal}, \text{buffered}\}$.
Ensure: Per-timestep binary alarms $\hat{y}(\tau) \in \{0,1\}$ and continuous score $N_{\text{tot}}(\tau)$.
Phase 1: Disjoint Training, Whitening & Adaptation ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q$)
Split $\mathbf{X}_{\text{nom}}$ chronologically into S-isolated ($\mathbf{X}_{\text{fit}}, \mathbf{X}_{\text{sel}}, \mathbf{X}_q, \mathbf{X}_{\text{cal}}, \mathbf{X}_{\text{hold}}$) and standardize channels using $\mathbf{X}_{\text{fit}}$ alone.
Set nested prefixes $S_{1:4} = \{16, 32, 48, 64\}$ (Full) or $S_{1:3} = \{16, 32, 64\}$ with $K \geq 25$ rFFT gate (Lite).
for stride-16 training batches in $\mathbf{X}_{\text{fit}}$ **do**
    Encode $z_{\text{ctx}} = E_\theta(\mathbf{x}_{\text{ctx}})$ and targets $z_{\text{tgt}}^{(m)} = E_\phi(\mathbf{x}_{\text{tgt}}^{(m)})$ with barycenter shift $-\bar{\boldsymbol{\mu}}_{\text{ctx}}$.
    Roll out sequential $\{P_{\text{sub}}, P_\theta\}$ (Full) or parallel single-LayerNorm predictor (Lite); update θ via (1) and $\phi \leftarrow m\phi + (1-m)\theta$.
end for
Select checkpoint on $\mathbf{X}_{\text{sel}}$ (or fixed epochs) and fit ($\bar{\boldsymbol{\mu}}_{\text{res}}, \boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$) on $\mathbf{X}_{\text{fit}}$.
if tier = Full and $q_{\text{wsp}} = 1$ **then**
    Rank drifting coordinates on ($\mathbf{X}_{\text{fit}}, \mathbf{X}_q$); run Steps Q1–Q4 for $k \in \{4, 2\}$ and keep edit iff $\mathbf{X}_q$ split and 0.92-quorum gates pass.
end if
Fit and freeze stabilization scales and blending weights $w_{m,c}$ (2) on $\mathbf{X}_q$.
Phase 2: Untouched Circuit-Breaker Calibration on $\mathbf{X}_{\text{cal}}$
Compute frozen stabilized residuals $\tilde{\mathbf{E}}_{\text{cal}}$ on untouched $\mathbf{X}_{\text{cal}}$ via (2).
if tier = Full **then**
    Expand $\tilde{\mathbf{E}}_{\text{cal}}$ into $7D = 224$ dials (position, drift, fluctuation, endpoint) and 32 right-edge onset relays.
    Fit MAD scales (m_j, s_j), hijack guards γ_j, tripwires τ_j (4), and quorums $(K_{224}, K_{128}, K_{\text{edge}})$ (5).
else ▷ Preferred Deployment Tier: Laminar-JEPA-Lite
    Extract $4D = 128$ position-plus-drift dials ($|\tilde{e}_{m,c}|, |\bar{e}_c|$), fit τ_j (4), and set single quorum $K_{\text{Lite}} = \max(1.12\, q_{0.995}, 3.5)$.
end if
Phase 3: Online Inference on $\mathbf{X}_{\text{hold}}$ or $\mathbf{X}_{\text{test}}$
for each stride-1 window $[t-C, t+S)$ **do**
    Compute $\tilde{\mathbf{E}}(t) \to \mathbf{d}(t)$ and binary trips $b_j(t) = \mathbb{I}(z_j(t) > \tau_j)$ (plus relays $r_c(t)$ if Full).
    Attribute active counts to right edge $\tau = t + S - 1$ (causal) or prefix spans $[t, t + S_m)$ (buffered).
    Emit $\hat{y}(\tau) \in \{0,1\}$ when active counts exceed $(K_{224}, K_{128}, K_{\text{edge}})$ (5) or K_{Lite}.
end for

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup, Two-Stage Nominal Calibration, and Statistical Design

All architectures use context $C = 256$, target horizon $S = 64$ ($W = 320$, or $S \in \{1, 4, 16, 32, 64\}$ in Section VII-C), latent dimension $D = 32$, nominal training stride 16, and evaluation stride 1. Models are trained with AdamW [52] (learning rate 10^{-3}, weight decay 10^{-4}) and zero per-dataset tuning: batch 16 for 15–20 epochs on development cohorts, and batch 16–64 for 10–12 epochs with gradient clipping (1.0) and cosine annealing on archive cohorts.

a) Disjoint Nominal Calibration Protocol, Boundary Isolation, and Cross-Fitting: Each stream's nominal history $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Table I) into a model-fitting prefix $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = T_{\text{nom}} - \lfloor 0.35\, T_{\text{nom}} \rfloor$) and a held-out nominal suffix $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$: (1) *Stage A (Disjoint weight fitting, checkpoint selection, and nominal adaptation):* Reusing calibration windows for early stopping, blending weights (2), or weight quenching can make nominal thresholds optimistic. All archive benchmarks ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$, Table VI) therefore

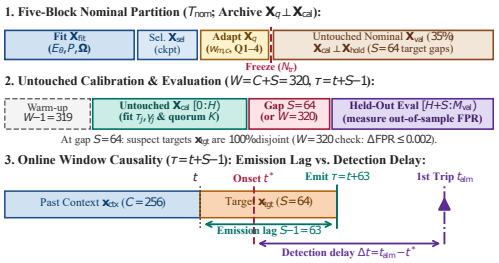


Fig. 2. Disjoint nominal evaluation protocol (C=256, S=64, W=320): (Row 1) five-block archive partition into S-isolated weight-fitting ($\mathbf{X}_{\text{fit}}$), checkpoint-selection ($\mathbf{X}_{\text{sel}}$), and nominal adaptation ($\mathbf{X}_q$) prefixes versus $\mathbf{X}_{\text{val}}$, (Row 2) untouched threshold calibration ($\mathbf{X}_{\text{cal}}$) and held-out nominal evaluation ($\mathbf{X}_{\text{hold}}$), and (Row 3) structural emission lag ($S-1$) versus empirical detection delay (Δt).

TABLE I
Nominal Partitioning and Calibration Protocol by Cohort

Cohort (Tables)	Weights (%)	Ckpt. Sel.	Adapt. / Quench	Thresholds
Dev. 19 Cache (IIa,b, III)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	None (unblended)	$\mathbf{X}_{\text{cal}}$
Dev. 33 Suite (IIc, IV, V)	$\mathbf{X}_{\text{tr}}$ (65)	Fixed ep.	Shared $\mathbf{X}_{\text{cal}}$	$\mathbf{X}_{\text{cal}}$
Dev. 19 4-Block (Sec. V-B)	$\mathbf{X}_{\text{fit}}$ (50)	Fixed ep.	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Archive 118/141 (VI)	$\mathbf{X}_{\text{fit}}$ (43)	$\mathbf{X}_{\text{sel}}$	Disjoint $\mathbf{X}_q$	$\mathbf{X}_{\text{cal}}$
Held-Out / Test ($\mathbf{X}_{\text{hold}}, \mathbf{X}_{\text{test}}$)	Frozen	Frozen	Frozen	Frozen

enforce a strictly disjoint *five-block nominal sequence* inside $\mathbf{X}_{\text{tr}}$: $\mathbf{X}_{\text{fit}}$ ($\approx 43\%$ of T_{nom}) fits channel standardization, encoder/predictor weights, running barycenter $\bar{\boldsymbol{\mu}}_{\text{ctx}}$, and precision $\boldsymbol{\Omega} \in \mathcal{S}_+^{MD}$; $\mathbf{X}_{\text{sel}}$ ($N_{\text{sel}} \geq 96$ windows) selects the early-stopping checkpoint; and $\mathbf{X}_q$ ($N_q \in [160, 2{,}048]$ stride-1 windows separated from $\mathbf{X}_{\text{sel}}$ and $\mathbf{X}_{\text{val}}$ by $S = 64$-step target-isolation gaps) fits and freezes (2) and optional split-gated quenching (Steps Q1–Q4) before $\mathbf{X}_{\text{cal}}$ is ever touched. Fixed-epoch development suites (Table I) omit $\mathbf{X}_{\text{sel}}$: the 19-stream residual cache skips (2) and quenching altogether, while a paired four-block check ($\mathbf{X}_{\text{fit}} \to \mathbf{X}_q \to \mathbf{X}_{\text{cal}} \to \mathbf{X}_{\text{hold}}$) on the 19 development streams confirms that disjoint $\mathbf{X}_q$ adaptation holds causal FPR_{hold} at 0.96% (0.58% median; 64.8% ER, 0.233 buffered F1) for *Laminar-JEPA-Lite* and 1.05% (0.78% median; 64.6% ER) for *Laminar-JEPA*, matching Table IV (1.0% and 1.3%) while trimming $\mathbf{X}_{\text{fit}}$ from 65% to 50% lowers short-stream causal F1 ($0.133 \to 0.071$ and $0.144 \to 0.076$). On short histories where carving out five blocks shrinks calibration support, *two-fold chronological cross-fitting* swaps $[0 : H)$ and $[H+S : M_{\text{val}})$ under consensus voting ($\hat{y}^{(1)} \wedge \hat{y}^{(2)}$), lowering 19-stream cache FPR_{test} from 1.50% to 0.92% (median $1.07\% \to 0.38\%$) at 66.7% median ER. (2) *Stage B (Untouched circuit-breaker calibration and boundary isolation on $\mathbf{X}_{val}$):* Stride-1 windows of length $W = 320$ inside $\mathbf{X}_{\text{val}}$ yield $M_{\text{val}} = N_{\text{val}} - W + 1$ valid window indices. Splitting at $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the untouched *nominal calibration block* $\mathbf{X}_{\text{cal}} = [0 : H)$ and *held-out nominal evaluation block* $\mathbf{X}_{\text{hold}} = [H+S : M_{\text{val}})$, separated by an $S = 64$-step isolation gap $[H : H+S)$ so calibration and evaluation share zero target observations (a full $G = 320$ window gap changes causal FPR by < 0.25 percentage points). On $\mathbf{X}_{\text{cal}}$ alone, the frozen model fits dial medians m_j, MAD scales s_j, hijack guards γ_j, tripwires τ_j (4), and quorums (K_{Lite} or $(K_{224}, K_{128}, K_{\text{edge}})$ in (5); τ_{SPOT} for 1D baselines), frozen on $\mathbf{X}_{\text{hold}}$ and $\mathbf{X}_{\text{test}}$. Raw window count $N_{\text{cal}} = H$ is distinguished from autocorrelation-adjusted *effective sample size* $N_{\text{eff},s} = N_{\text{cal}}/\tau_{\text{int},s}$ via stream- and statistic-specific Sokal initial-positive-sequence estimators (τ_{int} up to 206; Section VII-E).

b) Hierarchical Statistical Units, Dataset-Clustered Inference, and Hypothesis Testing: Because streams within multi-stream archives share hardware platforms, we report a two-level inference hierarchy after averaging seeds within each stream (seed standard deviation (SD)): (1) *Dataset-clustered and stream-level inference:* Because stream-level p-values treat same-spacecraft channels as independent, *dataset-clustered inference* (95% cluster-bootstrap CIs and exact 2^D dataset sign-flip permutation tests) is emphasized alongside stream-level paired Wilcoxon tests and 95% stream-bootstrap CIs (2,000 resamples). For multiplicity reporting on the 118 untouched streams, the focal causal contrasts comparing *Laminar-JEPA-Lite* against Multi-Horizon JEPA (1D) and TS-JEPA yield Benjamini–Hochberg (BH) [40] adjusted stream-level $p = 0.014$ and 3.6×10^{-4} on Point-F1 (raw $p = 0.014$ and 1.8×10^{-4}) and adjusted dataset sign-flip $p = 0.0078$ and 0.012 on ER (raw $p = 0.0039$ and 0.012, positive on $9/9$ and $8/9$ datasets); other quoted p-values are raw and descriptive, and non-Exathlon subsets ($n = 101$, 8 datasets) verify sensitivity to archive composition. (2) *12-fold LODO audit:* Across the 12 LODO folds on the 141-stream archive, *Laminar-JEPA-Lite* retains positive gains in all $12/12$ folds on causal ER ($\Delta \in [+0.099, +0.141]$ vs. Multi-Horizon 1D), buffered Point-F1 ($\Delta \in [+0.002, +0.017]$), and causal Point-F1 vs. TS-JEPA ($\Delta \in [+0.013, +0.027]$, $\max p = 0.0054$), and in $9/12$ folds on causal Point-F1 vs. Multi-Horizon 1D (Section VII-E).

B. Consolidated Cohort Organization, Development vs. Held-Out Separation, and Baseline Fairness

Across all evaluations, we study a two-tier family of multi-horizon coordinate circuit-breaker architectures: *Laminar-JEPA-Lite* is the preferred deployment configuration (parallel three-horizon 128-dial position-plus-drift panel, single quorum, zero weight editing), while *Laminar-JEPA* is retained as the expressive four-horizon 224-dial reference variant with optional split-gated quenching (*Laminar-JEPA (w/o Quench)* omits Steps Q1–Q4). Both causal trailing and retrospective buffered attribution are paired on identical splits within each cohort (aside from a single-horizon 130-stream buffered baseline screen in Section VII-A): (1) *Development cohorts (design-inspected, Table I):* The 19-stream residual cache (6 datasets) evaluates the dial ladder, component ablations, and conformal/statistical-process-control (SPC) regimes on identical 4H unquenched residuals (Table II(a,b), Table III); the 33-stream development benchmark (7 datasets: 19 non-GHL streams in Table II(c) plus 14 GHL streams in Table IV, paired with a four-block $\mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$ sensitivity run) compares both tiers against 1D JEPAs and observation-space baselines; and tripwire/quenching constants were fixed on 23 tuning streams (19 non-GHL plus `SMAP T-2`, `MSL P-10`, `Daphnet S09R01E2`, and `OPPORTUNITY S1-ADL3`). (2) *Untouched held-out cohorts (primary generalization evidence):* Excluding those 23 design streams from the 141-stream archive (12 datasets, strictly disjoint $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$) leaves 118 **untouched held-out streams** across 9 datasets (101 non-Exathlon streams across 8 datasets; Table VI), supplemented by a single-seed 4H Q1–Q4 step-isolation audit on the same 118 streams (Section VI-A). None of the 118 held-out streams was inspected during architecture or threshold design.

Standardized Baseline Fairness Protocol: Scalar baselines (TS-JEPA, Multi-Horizon JEPA (1D), TCN Forecaster/Autoencoder, PCA Forecaster, Isolation Forest, TimesNet [5], TranAD [6], Anomaly Transformer [7], NCAD [10], and DCdetector [16]) share the identical chronological splits, training-only standardization, isolation gaps, and SPOT calibration ($u = p_{98}$, $q = 10^{-3}$), whereas conformal, SPC, and simpler coordinate detector-bank baselines in Tables III and VI share the identical multi-horizon whitened residuals and nominal calibration alert rate. *Laminar-JEPA-Lite* uses 141.9k trainable parameters at $K < 25$ and 222.3k at $K \geq 25$ versus 295.8k for *Laminar-JEPA* (206k–649k across external neural baselines; circuit-breaker panels add zero trainable weights). Runs used PyTorch 2.4 / CUDA 12.4 on an RTX 5090 (0.15 ms/window batched inference for *Laminar-JEPA-Lite* vs. 0.38 ms/window for *Laminar-JEPA*, 2.43× speedup; 188 MB peak memory); code: https://github.com/quaid281/NCAD.

C. Multi-Tier Operational Evaluation Metrics and Aggregation

To prevent point-adjustment inflation [41] and aggregation ambiguity, all binary false-alarm and Point-F1 metrics are computed *per valid timestep* τ (never point-adjusted and never per-window without timeline attribution): FPR_{hold} averages $\hat{y}(\tau)$ over valid timesteps in $[H + S : M_{\text{val}})$, FPR_{test} averages $\hat{y}(\tau)$ over nominal test timesteps ($y(\tau) = 0$), and Point-F1, Precision, Recall, and PR-AUC (from $N_{\text{tot}}(\tau)$) score all valid test timesteps against $y(\tau) \in \{0, 1\}$. Event-level ER measures the fraction of contiguous $y = 1$ segments with ≥ 1 alarm, and delay $\Delta t = t_{\text{alm}} - t^*$ is the step lag from segment onset t^* to first in-segment alarm. Unless explicitly labeled *median* or *Dataset-Macro*, cohort headline numbers report *stream-level arithmetic means* of within-stream seed averages, paired with stream medians for skewed FPR distributions and non-deployable post-hoc test-FPR controls ($f^* \in \{0.01, 0.05\}$) strictly for diagnostic comparison.

VI. Ablation Studies and Analysis

A. Ablation 1: Perspective Expansion Ladder, Circuit-Breaker Components, and Weight-Space Quenching

Table II(a),(b) and Fig. 3(a) evaluate perspective expansion and panel components on the 19-stream residual cache (not paired to the separately trained run in Panel (c)):

1) *Reference 224-dial panel vs. distilled position-plus-drift subsets:* We retain 224-dial *Laminar-JEPA* as the full-perspective reference architecture because evaluating all four perspectives on one residual tensor isolates which views aid real-time streaming versus retrospective attribution. Although 224 dials lift causal Point-F1 from 0.013 to 0.071 over 1D scoring ($p = 0.0023$; buffered $+0.033$, $p = 0.53$) and lead buffered Point-F1 (0.224 vs. 0.198), pruning fluctuation, endpoint, and relay switches down to the 160-dial position-plus-drift subset (128 sub-horizons + 32 horizon means, single quorum) raises causal Point-F1

TABLE II
CIRCUIT-BREAKER DIAL EXPANSION, COMPONENT ABLATIONS, AND MATCHED-BACKBONE STUDY

(a) Perspective Expansion Ladder (1 → 224 Dials, n=19)

Perspective Expansion Stage	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
1D Scalar Collapse (1 Dial)	.013	.191	**.315**	.413	.459	**.550**	166.2	143.1
32-Bit Full Horizon (m=4)	.050	.135	.255	.381	.500	.243	123.4	65.5
128-Bit Sub-Horizons (m=1..4)	.063	.216	.294	**.449**	.556	.462	110.6	**38.8**
160-Bit (+32 Horizon Drift)	**.098**	.198	.304	.429	**.591**	**.550**	**74.8**	108.3
192-Bit (+32 Horizon Fluc.)	.067	.192	.299	.443	.573	.453	79.1	101.6
224-Bit (+32 Endpoint/Relays)	.071*	**.224**	.299	.448	.564	.506	90.1	83.7

* Versus 1D, causal F1 difference $+0.057$ (stream-bootstrap 95% CI $[0.016, 0.120]$; Wilcoxon $p = 0.0023$, 12/2 wins/losses). Buffered difference $+0.033$ (CI $[-0.017, 0.093]$, $p = 0.53$). The scalar spectral-modulation ablation in the separately trained study remains null.

(b) Circuit-Breaker Component & Quorum Ablation (n=19)

Component / Quorum Variant	Pt-F1 ↑ Caus.	Pt-F1 ↑ Buf.	PR-AUC ↑ Caus.	PR-AUC ↑ Buf.	ER ↑ Caus.	ER ↑ Buf.	Delay Δt ↓ Caus.	Delay Δt ↓ Buf.
224-Bit Breaker (Full Panel)	.071	.224	.299	.448	**.565**	**.506**	90.1	83.7
224-Bit Strict (High Quorum K_{strict})	.047	.168	.299	.448	.494	.439	207.6	**54.3**
w/o 32 Right-Edge Relays	.068	.225	.299	**.452**	**.565**	**.506**	90.7	83.7
w/o Prefix Routing (Unif. Smear)	.071	.222	.299	.440	**.565**	.497	90.1	115.4
w/o Hijack Guards (γ_j=1)	**.073**	**.226**	**.299**	.448	**.565**	**.506**	**70.1**	83.3

Panels (a,b): 19 streams × 3 seeds on the same whitened-residual cache, without the stabilized blend in (2). Panel (c) uses a separate training run; cross-panel comparisons are not paired.

(c) Matched-Backbone Architectural Comparison (n=19 Development Streams)

Architecture Arm	Tail / Drift γ_2	Tail / Drift D_{KS}	Held-Out FPR Caus.	Held-Out FPR Buf.	Point-F1 Caus.	Point-F1 Buf.	PR-AUC Caus.	PR-AUC Buf.	Causal Event Recall Auto.	Causal Event Recall q^*=0.05	Causal Event Recall f^*=0.01	Causal Event Recall f^*=0.05
Laminar-JEPA (224-Bit)	0.92	0.190	0.013	0.128	0.144	**0.244**	**0.311**	**0.407**	0.673	0.795	0.612	0.757
Laminar-JEPA (w/o Quench)	0.88	0.190	0.016	0.128	0.142	0.235	0.304	0.381	**0.695**	0.815	**0.631**	0.765
Laminar-JEPA-Lite (128-Bit)	2.33	0.182	0.010	0.061	0.133	0.242	0.301	0.397	0.650	0.811	0.560	**0.772**
TS-JEPA (Unconstrained)	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
Multi-Horizon JEPA (1D)	2.30	0.179	**0.006**	0.078	0.050	0.148	0.259	0.331	0.478	0.752	0.562	0.698
Multi-Horizon JEPA (α_{LT}=0, 1D)	2.19	0.183	**0.006**	0.083	0.055	0.149	0.261	0.330	0.439	0.755	0.578	0.711
JEPA Context Autoencoder	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
JEPA Frozen Target	1.16	**0.167**	0.009	0.098	0.146	0.235	0.273	0.328	0.503	**0.816**	0.398	0.600
TS-JEPA (ℓ_2 Scoring)	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
PCA Linear Forecaster (LN)	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
TCN Obs. Forecaster (LN)	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
Isolation Forest	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
TCN Obs. Forecaster	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
TCN Obs. Autoencoder	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
TimesNet	5.54	0.438	0.243	0.300	**0.209**	0.197	0.281	0.329	0.588	0.618	0.217	0.407
TranAD	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

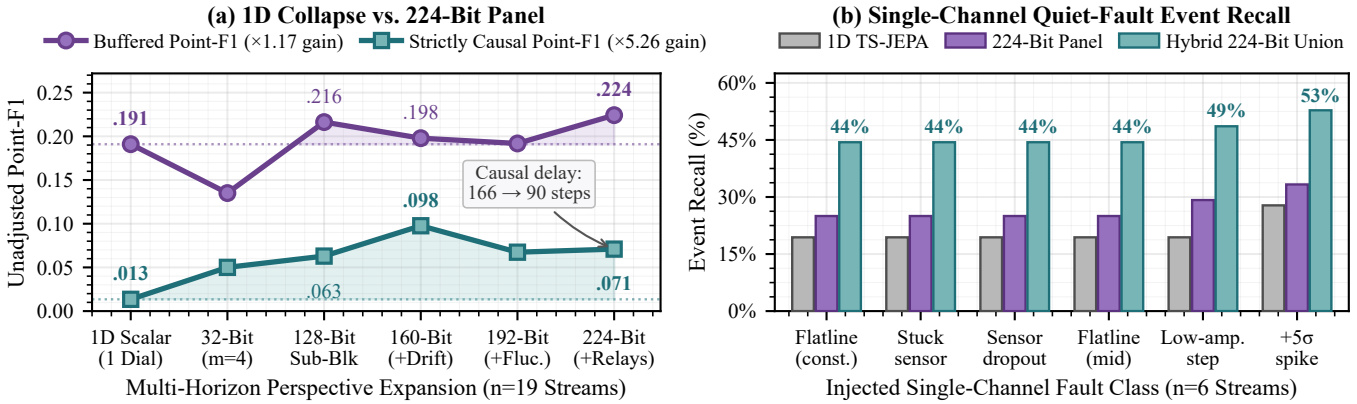


Fig. 3. Development studies. **(a)** Causal and buffered Point-F1 across the six-step expansion on cached whitened residuals from 19 streams (three seeds; no stabilization blend). **(b)** Event recall for injected single-channel faults in a separate six-stream study, comparing 1D TS-JEPA, 224-Bit Breaker, and their channel-level union.

to 0.098 and ER to 59.1%, cuts delay from 90.1 to 74.8 steps, and lowers nominal FPR (0.55% held-out, 0.87% test vs. 0.76%, 1.50%) at 29% fewer dials, motivating *Laminar-JEPA-Lite* as the preferred deployment tier. Scalar R_{LT} modulation remains null in Panel (c).

2) *Relays, PR-AUC score weighting, quorum rules, and hijack guards:* Removing the 32 onset relays lowers causal Point-F1 (0.071 → 0.068), and replacing the three-counter quorum (5) with one 224-dial quorum lowers causal Point-F1 to 0.063 ($\text{FPR}_{\text{hold}} = 0.0055$; Table III). Replacing the double-position continuous score $N_{\text{tot}} = N_{224} + N_{128} + 2N_{\text{edge}}$ with unweighted $N_{224}+N_{\text{edge}}$ or N_{224} leaves binary alarms identical and changes PR-AUC by at most 0.004 (0.299 → 0.297 causal, 0.448 → 0.444/0.446 buffered; *Laminar-JEPA-Lite* uses unweighted N_{128}). Removing hijack guards ($\gamma_j = 1$) raises causal Point-F1 to 0.073 (70.1-step delay) on this cache when nominal drift is mild; none removes the $S-1$ reporting lag.

3) *Quenching step isolation and LayerNorm coupling:* On single-seed 4H checkpoints across all 118 untouched streams (unquenched causal $\text{FPR}_{\text{hold}} = 6.76\%$, F1 = 0.0901, ER = 63.8%), $k = 4$ predictor row damping (Step Q1) shifts unselected coordinates by a median 11.1% of nominal RMS (45.5% shifting > 10%) via 32D LayerNorm coupling without lowering FPR_{hold} (7.06%), while horizon-embedding shifts lower raw RMS on 68.2% of edited coordinates (4.1% spillover). Buffer recentering and precision decoupling (Steps Q2–Q3) lower FPR_{hold} to 6.71% at lower causal F1 (0.0826), whereas adding the split gate (Step Q4, accepted on 39/118 streams) reaches 6.62% while preserving causal F1 (0.0904). Because ungated edits hurt $k = 2$ causal F1 on 11/19 development streams (0.141 → 0.049) and $k \in \{8, 12\}$ cuts PR-AUC (0.292 → 0.223), *Laminar-JEPA-Lite* omits weight editing altogether.

TABLE III
COORDINATE DETECTOR-BANK REGIMES ON CACHED RESIDUALS

Detector & Calibration Regime	Pt-F1 ↑	ER ↑	$\mathrm{FPR}_{\mathrm{hold}}$ ↓	$\mathrm{FPR}_{\mathrm{test}}$ ↓
Multi-horizon circuit-breaker panels (nominal quorum)				
160-dial pos.+drift (single quorum)	**0.098**	0.591	0.0055	0.0087
224-dial panel (deployed)	0.071	0.564	0.0076	0.0150
224-dial single quorum (no relay)	0.063	0.564	0.0055	0.0088
1. Per-coordinate marginal union (no quorum, 128 coords)				
Per-coord. $q_{.985}$ union	0.291	0.913	0.6651	0.7353
Per-coord. .0034 marginal union	0.303	0.874	0.4057	0.4765
2. Family-wise / FDR multiplicity control (α, q=.005)				
Per-coord. conformal (Šidák)	0.000	0.000	**0.0000**	**0.0000**
224-dial conformal BH FDR	0.000	0.026	**0.0000**	**0.0000**
3. System-level matched nominal rate (mean .0034 on $\mathbf{X}_{cal}$)				
Raw 128-coord. top-k vote	0.068	0.564	0.0052	0.0096
Continuous 128-coord. $\max_j z_j$	0.056	0.658	0.0078	0.0147
Discrete conformal $\min_j p_j$	0.265	0.843	0.2111	0.2831
Multivar. conf. ($\|\mathbf{e}_{\mathrm{white}}\|_2^2$)	0.072	0.608	0.0062	0.0095
Fisher p-pool (224 dials)	0.109	0.643	0.0105	0.0183
Maximum 224-dial z	0.083	0.652	0.0093	0.0157
Hotelling T^2 (recalibrated)	0.283	0.816	0.2900	0.4831

19 streams × 3 seeds, causal means. System-matched rules match each stream's nominal panel alert rate after aggregation (Stouffer .095 F1, .0087 $\mathrm{FPR}_{\mathrm{hold}}$; 224-dial top-$k$ .070, .0077; Mahalanobis conf. .2789, CUSUM .3457, MEWMA .5862 $\mathrm{FPR}_{\mathrm{hold}}$).

Marginal, multiplicity-controlled, and system-matched detector banks: Evaluating coordinate detector banks fairly requires separating three calibration regimes on the identical 19-stream whitened residuals without test-label tuning (Table III): (1) *Per-coordinate marginal union:* Thresholding 128 coordinates individually at $q_{0.985}$ or even at the panel's 0.34% rate without a quorum compounds false alarms across coordinates ($\mathrm{FPR}_{\mathrm{hold}} = 66.5\%$ and 40.6%). (2) *Family-wise and false discovery rate (FDR) multiplicity control:* Applying Šidák (0.5%) across 128 coordinates or BH FDR ($q = 0.005$) across 224 dials hits the discrete conformal p-value floor $(N_{\mathrm{cal}}+1)^{-1} > 3.9\times10^{-5}$, paralyzing causal recall (F1 ≤ 0.001). (3) *System-level matched nominal alarm rate:* Calibrating the aggregated statistic directly to each stream's nominal panel rate (0.34%) shows that discrete conformal $\min_j p_j$ still jumps to $\mathrm{FPR}_{\mathrm{hold}} = 21.1\%$ out-of-sample due to boundary ties whenever any coordinate exceeds its calibration maximum, and recalibrated Mahalanobis conformal, Hotelling T^2, multivariate exponentially weighted moving average (MEWMA), and cumulative sum (CUSUM) charts inflate to 27.9%–58.6%. Among stable system-matched rules, the 160-dial position-plus-drift panel (F1 $= 0.098$, $\mathrm{FPR}_{\mathrm{hold}} = 0.55\%$) outperforms unguarded 128-coordinate top-k voting (0.068) and continuous 128-coordinate $\max_j z_j$ (0.056) while matching Fisher/Stouffer p-pooling ($0.109/0.095$ at $1.05\%/0.87\%$ $\mathrm{FPR}_{\mathrm{hold}}$) and training-whitened ℓ_2^2 conformal (0.072), confirming that horizon-drift expansion and hijack-guarded quorum voting improve over raw coordinate banks.

B. Ablation 2: Single-Channel Faults and Complementary Channel-Level Anomaly Branch

Even when flatlining shifts the normalized summary (0.64 vs. 0.77 relative divergence for a variance-matched harmonic flip on `SMAP P-3`), terminal LayerNorm spreads static error thinly across all 32 coordinates (12.5 vs. 62.1 switches; 6.3% vs. 97.7% window alerts), and single-sensor faults suffer further $O(1/K)$ dilution across $K-1$ nominal channels. Extending Fig. 3(b) (n=6), we evaluate a causal channel branch across all 19 multivariate development streams ($K \in [5, 55]$) without per-stream or test-label tuning. Per channel, three 64-step rolling statistics—log-variance drop $-\ln(\mathrm{Var}_{64}(x_c) + 10^{-4})$, training-fit first-order autoregressive (AR(1)) residual RMS, and mean shift $|\bar{x}_{64,c} - \mu_c|/\sigma_c$—are thresholded at $1.25\, q_{0.999}$ on nominal validation windows (floor 6.0 MAD, $1.25\times$ half-split drift multiplier) with a 24-of-40-step persistence gate. On the highest-variance active channel $c^\star = \arg\max_{c:\sigma_c^{\mathrm{tr}}>10^{-8}} \sigma_c^{\mathrm{tr}}$, a deterministic 9-point grid over $[320, N_{\mathrm{test}} - 520]$ (stride-50 fallback) places three nonoverlapping $L_{\mathrm{inj}} = 200$-step faults (57 events/class, gap ≥ 264) inside clean segments ($y^{\mathrm{orig}} \equiv 0$ on $[s-64, s+264)$). For $s \leq t < s+200$ using training $(\mu_c, \sigma_c, q_{.995,c})$:

$$x'_{t,c^\star} = \begin{cases} \mu_{c^\star} & \text{variance collapse,} \\ \mu_{c^\star} + 1.35(-1)^{t-s}\sigma_{c^\star} & \text{prediction shock,} \\ x_{s-1,c^\star} & \text{sensor freezing,} \\ \max(q_{.995,c^\star}, \mu_{c^\star} + 2.2\sigma_{c^\star}) & \text{saturation,} \\ x_{t,c^\star} + 2.5\sigma_{c^\star}(t-s)/199 & \text{gradual degradation,} \\ x_{t,c^\star} + 1.5\sigma_{c^\star} & \text{isolated bias.} \end{cases} \tag{6}$$

Outside $[s, s+200)$, all channels remain unchanged. Injected-event recall rises from 63.2% (standalone panel) to 93.0% for variance collapse, 63.2% to 94.7% for freezing, and 82.5% to 94.7% (96.5% full branch) for saturation under the variance-collapse union; the full channel branch raises recall from 75.4% to 96.5% on prediction shocks, 59.6% to 80.7% on gradual degradation, and 68.4% to 86.0% on isolated bias. However, on the *unmodified* test streams, causal $\mathrm{FPR}_{\mathrm{test}}$ rises from 1.5% (standalone panel, F1 $= 0.071$) to 6.8% (variance-collapse union, F1 $= 0.070$) and 19.4% (full branch, F1 $= 0.129$). Synthetic single-channel injections do not establish detection on natural sensor failures, and the full union's 19.4% false-alarm rate makes it unsuitable as a default detector.

C. Ablation 3: Multiscale Dyadic-Shell Front-End vs. Standard Convolutional Stem

Retraining the multi-horizon JEPA backbone with and without 4-shell rFFT splitting on five streams (`SMAP P-3`, `MSL M-1`, `SMD machine-1-2`, `Daphnet S01R01E1`, `GECCO`; three seeds) reveals a clear channel-count threshold: shell splitting improves high-channel spacecraft detection (K=55 `MSL M-1` buffered F1 $0.085 \rightarrow 0.410$; K=25 `SMAP P-3` causal F1 $0.212 \rightarrow 0.321$) by separating thermal drift from actuator chatter, but lowers F1 at K=9 (`Daphnet`, `GECCO`), setting the $K \geq 25$ spectral gate in *Laminar-JEPA-Lite*.

D. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing all 16 matched-budget arms in Table II(c) isolates five mechanisms: (1) raw-observation forecasting and autoencoding share heavy tails ($\gamma_2 \approx 6.2$), large drift ($D_{\mathrm{KS}} \approx 0.45$), and $\approx$ 25% causal FPR; (2) a D=32 LayerNorm bottleneck in observation space cuts kurtosis ($\gamma_2 = 0.91$) and halves FPR, yet raw targets keep drift high ($D_{\mathrm{KS}} = 0.376$); (3) learned D=32 latent targets and whitening halve drift ($D_{\mathrm{KS}} = 0.214$ in TS-JEPA and 0.179–0.190 across multi-horizon JEPA tiers, $p \leq 0.001$); (4) linear PCA and Isolation Forest keep low FPR but miss nonlinear faults; and (5) frozen

targets and ℓ_2 scoring retain $< 2\%$ causal FPR, whereas latent context autoencoding lowers causal F1.

VII. Results and Discussion

A. Development 33-Stream and Untouched Held-Out 118/141-Stream Benchmark Results

Following the cohort hierarchy in Section V-B, Table IV reports the 33-stream development benchmark under retrospective buffered attribution (Panel (a)), causal streaming (Panel (b)), and per-dataset stratification (Panel (c)), while Table VI reports the strictly disjoint five-block ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$) benchmark across all 118 untouched held-out streams and the 141-stream archive. On a separate 130-stream baseline screen, TS-JEPA reaches stream-mean Point-F1 0.186 versus TranAD 0.152, TimesNet 0.151, DCdetector 0.097, NCAD 0.088, and Anomaly Transformer 0.082. The observed reduction in nominal false alarms with latent scoring is specific to these protocols; spherical normalization alone does not prove a distributional guarantee.

Operational trade-offs and configuration guidance: On the 33-stream benchmark, the 224-dial reference variant *Laminar-JEPA* edges out *Laminar-JEPA-Lite* on unweighted buffered Point-F1 (0.244 vs. 0.242) and causal ER (67.3% vs. 65.0%), whereas *Laminar-JEPA-Lite* halves buffered FPR_{hold} (6.1% vs. 12.8%), leads 7-dataset macro Point-F1 (0.171 vs. 0.133), and cuts parameters by 35%–52% and inference latency by $2.43\times$. In practice, *Laminar-JEPA-Lite* is our preferred deployment configuration for streaming and buffered monitoring of abrupt and intermediate-onset faults, *Laminar-JEPA* is retained as the full-perspective reference variant for diagnostic ablation and peak development buffered F1, short horizons ($S \leq 4$) address sub-window reporting lag, and raw-level variance or trend charts should be paired only when targeting single-sensor freezes or slow monotonic process ramps.

Scope boundaries and controlled slow-ramp sweep: Observation-space reconstructors show high nominal false-alarm rates here, whereas learned latent targets lower false alarms without guaranteeing light tails on arbitrary data. Conversely, because the encoder summarizes only the preceding $C = 256$ steps, gradual drift that shifts negligibly across one context window causes context and target summaries to co-move, explaining the 0.000 Point-F1 on the 14 GHL slow-ramp streams (low false alarms without recall is not detection). To map this boundary, we injected linear ramps of amplitude $a \in \{0.5, 1, 2, 4\}\sigma$ and duration $L \in \{64, 256, 1024, 4096\}$ (slopes a/L from 6.25×10^{-2} to $1.22\times10^{-4}\sigma$/step, plus a 256-step plateau; 15 events/cell across 10 clean test segments on $\lceil 0.3K \rceil$-channel and 1-channel subsets). For steep multi-channel ramps ($L = 64$, $a \in [1, 4]\sigma$), the 224-bit panel reaches 86.7%–88.7% causal detection probability at 46–54-step median delay and 1.7%–1.9% false-alarm rate ($+5.7$–7.7 percentage points above the clean-segment Poisson episode baseline; $+19.4$–24.1 points for 1D SPOT at 60–67-step delay). By $L = 256$ ($= C$), panel excess causal detection narrows to $+0.5$–2.5 points (50–71-step delay), and for $L \geq 1024$ (slope $\leq 3.9\times10^{-3}\sigma$/step) raw hit rates ($87\%$–$91\%$) reflect background episodes over 1,280–4,352-step spans, with chance-corrected detection falling to $[-4.8, +0.8]$ points across all JEPA detectors ($[-5.9, -1.0]$ on single-channel ramps). Fixed-window JEPAs thus target abrupt to medium-onset faults rather than gradual industrial drift.

B. Real-Time Streaming vs. Retrospective Buffered Attribution

Comparing Panels (a) and (b) of Table IV shows why 1D latent detectors degrade under real-time causal streaming. Buffered attribution averages residuals across overlapping predictions covering each timestamp ($\gamma_2 = 1.25$), whereas causal streaming evaluates a single right-edge step without temporal averaging ($\gamma_2 = 0.92$–2.33). Summing whitened coordinates into a 1D score allows one or two drifting coordinates to inflate the global SPOT threshold onto that heavier tail, missing sustained faults. Replacing the scalar sum with independently calibrated coordinate breakers isolates drifting coordinates. Against a rate-matched Bernoulli alerter $\text{ER}_{\text{rand}}(p) = E^{-1}\sum_{e=1}^{E}[1-(1-p)^{L_e}]$, random guessing catches only 28.1% of events at our $\leq 1.3\%$ causal FPR, whereas at TimesNet's 24.3% causal FPR, coin flipping catches 60.0% of events by chance alone, exceeding TimesNet's 58.8% recall.

C. Multi-Horizon Scaling Across $S \in \{1, 4, 16, 32, 64\}$ and Latency Trade-Offs

Because a target window of length S emits its causal decision at $\tau = t + S - 1$, $S = 64$ imposes a 63-step reporting lag. We retrain on five selected streams at $S \in \{1, 4, 16, 32, 64\}$ to examine the latency trade-off. At $S = 1$, all predictor states are compared with the same one-step target, so this setting has no distinct target-horizon decomposition; at $S = 4$, the shortest encoded prefix is two steps. In the buffered $S = 1$ setting, endpoint-difference votes have no routed span. These short-horizon results therefore test the implemented detectors, not distinct horizon views.

In Fig. 4(a), both *Laminar-JEPA* and *Laminar-JEPA-Lite* exceed 1D Multi-Horizon SPOT across all five horizons while holding test FPR below 3.6% ($\leq 2.4\%$ for *Laminar-JEPA-Lite*, held-out FPR $\leq 1.2\%$, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). At $S = 1$ and $S = 4$, *Laminar-JEPA* reaches $2.6\times$ and $4.3\times$ the scalar causal Point-F1 (0.037 and 0.050 vs. 0.014 and 0.012) with 77%–84% ER, while *Laminar-JEPA-Lite* reaches $2.1\times$ and $3.1\times$ (0.030 and 0.036) with 64%–82% ER; at $S = 4$, detection delay drops to 36.7–44.1 steps. Buffered Point-F1 peaks at 0.290 for *Laminar-JEPA* ($S = 32$) and 0.255 for *Laminar-JEPA-Lite* ($S = 64$) versus 0.182 for 1D SPOT. Five selected streams cannot establish a deployment rule across sampling rates or datasets.

D. Operating Budgets and Contextual vs. Point Anomalies

Fig. 4(b) and Table V examine detection across operating budgets and fault taxonomies:

1) *Autonomous vs. diagnostic operating budgets (Fig. 4(b), Table II(c)):* Along the causal recall–FPR frontier, the autonomous points of *Laminar-JEPA* and *Laminar-JEPA-Lite* occupy the $\leq 1.3\%$ held-out FPR regime without

TABLE IV
PRIMARY 33-STREAM DEVELOPMENT BENCHMARK ACROSS ATTRIBUTION REGIMES

	(a) Buffered Attribution (n=19 Non-GHL)					(b) Causal Attribution (n=19 Non-GHL)				GHL (n=14)		Macro
Architecture	**FPR ↓**	**Pt-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Pt-F1 ↑**	**DS-F1 ↑**
Multi-horizon coordinate circuit-breaker architectures												
Laminar-JEPA (224-Bit)	0.128	**0.244**	**0.407**	**0.669**	432.9	0.013	0.144	**0.673**	249.9	0.004	0.000	0.133
Laminar-JEPA-Lite (128-Bit)	0.061	0.242	0.397	0.636	59.3	0.010	0.133	0.650	73.8	0.005	0.000	**0.171**
1D scalar-scored latent predictive baselines (JEPAs)												
TS-JEPA (unconstrained 1D)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Multi-Horizon JEPA (1D)	0.078	0.148	0.331	0.430	127.1	**0.006**	0.050	0.478	154.0	**0.002**	0.000	0.073
Multi-Horizon JEPA (α_{LT}=0, 1D)	0.083	0.149	0.330	0.462	113.3	**0.006**	0.055	0.439	138.4	**0.002**	0.000	0.074
Observation-space reconstructive and linear baselines												
TimesNet	0.300	0.197	0.329	0.583	**48.5**	0.243	**0.209**	0.588	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Dataset Summary					Buffered Point-F1 ↑					Held-Out FPR (Causal / Buffered) ↓				
Dataset	n	K	E	π	$F1_{all}$	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**	**Lam.**	**Lite**	**TS-JEPA**	**TimesNet**	**TranAD**
Daphnet	6	9	72	0.148	0.256	0.214	0.179	0.145	**0.229**	0.144	**.009/.061**	.012/.091	.030/.091	.291/.313	.250/.298
GHL	14	16	26	0.013	0.026	0.000	0.000	0.000	**0.022**	0.020	**.003/.004**	.005/.005	.004/.004	.056/.061	.040/.046
Genesis	1	18	2	0.003	0.006	0.078	**0.191**	0.000	0.000	0.000	.009/.104	.008/.020	.004/.017	**.000/.000**	**.000/.000**
MSL	4	55	6	0.255	0.381	**0.284**	0.210	0.161	0.158	0.151	.013/.151	**.004/.049**	.054/.095	.161/.133	.132/.241
SMAP	6	25	7	0.150	0.236	0.357	0.375	**0.379**	0.289	0.167	.021/.223	.016/**.066**	**.014**/.135	.333/.549	.360/.513
Metro	1	5	25	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	.010/.014	.001/**.000**	.008/.014	**.000/.000**	**.000/.000**
SWAN	1	38	6615	0.279	0.437	0.000	**0.240**	0.119	0.000	0.000	**.000/.000**	.001/**.000**	**.000/.000**	**.000/.000**	**.000/.000**

Note: 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), trained separately from Table II(a,b); best in **bold**, second underlined. In (c), n = streams, K = channels, E = events, π = anomaly rate, $F1_{all} = 2\pi/(1+\pi)$, Lam. = Laminar-JEPA, Lite = Laminar-JEPA-Lite. Unweighted Laminar-JEPA Δt reflects one late SWAN trigger (0.0002 recall; median Δt is 50.5 buffered / 45.8 causal, or 54.8 / 57.2 ex-SWAN). 95% stream-bootstrap CIs (n=19): Lite buffered FPR [.030, .103], F1 [.151, .335] and causal FPR [.006, .014], F1 [.055, .236]; Lam. buffered F1 [.150, .357] and causal FPR [.009, .018], F1 [.058, .257] (vs. Multi-Horizon 1D [.013, .097] and TimesNet causal FPR [.123, .375]).

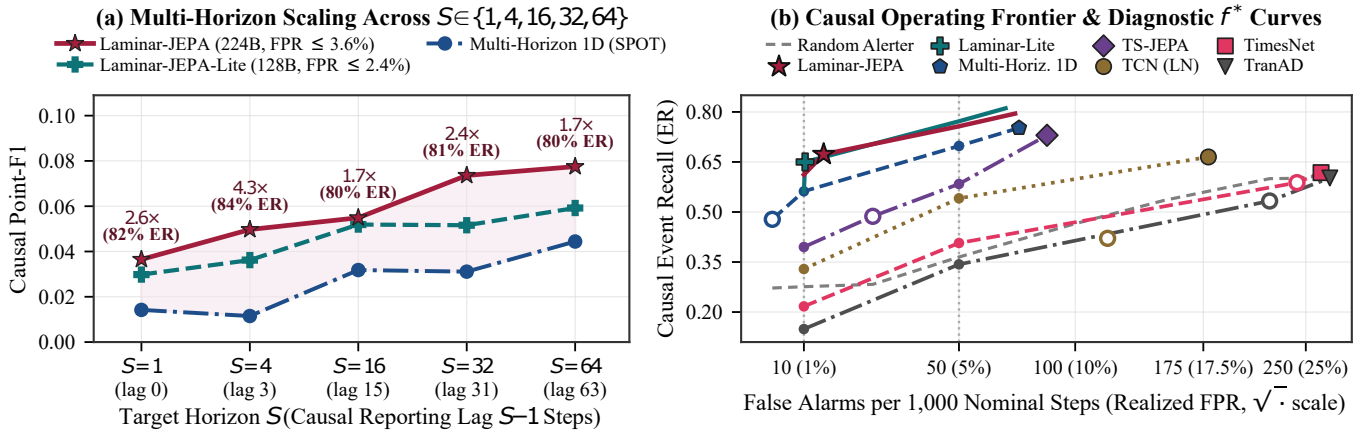


Fig. 4. Multi-horizon scaling ($S \in \{1, 4, 16, 32, 64\}$, causal reporting lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, n=5 streams) and causal operating frontier (n=19 streams). **(a)** Causal Point-F1 and event recall (ER) across S comparing *Laminar-JEPA (224B)*, *Laminar-JEPA-Lite (128B)*, and Multi-Horizon 1D SPOT (nominal FPR $\leq 3.6\%$ across all S, vs. $2.5\% \rightarrow 21.3\%$ for TimesNet). **(b)** Causal ER vs. realized false alarms ($\sqrt{\cdot}$ scale) across diagnostic budgets $f^* \in \{1\%, 5\%\}$ (dotted guides), autonomous points (*Laminar-JEPA* star, *Laminar-JEPA-Lite* cross, 1D open circles), and quantiles $q^* \leq 0.05$.

test-label tuning. At post-hoc diagnostic test-FPR controls ($f^* \in \{1\%, 5\%\}$), *Laminar-JEPA* and *Laminar-JEPA-Lite* catch 61.2% (63.1% unquenched) and 56.0% of events at $f^* = 1\%$ versus 39.5% for TS-JEPA, 21.7% for TimesNet, and 14.9% for TranAD.

2) *In-range contextual vs. point faults (Table V):* Under NASA's spacecraft taxonomy [4] (Panel (a)) and a 5σ amplitude split across all 19 development streams (Panel (b)), out-of-range point spikes ($> 5\sigma$) are flagged by reconstructors and amplitude rules alike, whereas in-range contextual faults decouple cross-channel phase or velocity inside normal amplitude bounds: on NASA contextual sequences, *Laminar-JEPA-Lite* and *Laminar-JEPA* achieve 87.5% and 77.5% autonomous ER at 1.3% and 2.2% FPR (84.0% and 80.0% at $f^* = 1\%$; 0.532 and 0.519 buffered PR-AUC) versus 62.5% at 41.5% FPR (56.0% at $f^* = 1\%$, 0.317 PR-AUC) for TimesNet.

E. Untouched Held-Out Generalization, Calibration Sample-Size Reliability, and Drift Decomposition

Table VI and archive-wide audits establish three generalization and calibration findings:

1) *Independent generalization under five-block non-leakage (Table VI):* Across the 118 untouched streams ($\mathbf{X}_{sel} \perp \mathbf{X}_q \perp \mathbf{X}_{cal}$, three seeds), *Laminar-JEPA-Lite* raises causal Point-F1 from 0.076 (Multi-Horizon 1D) and 0.057 (TS-JEPA) to 0.079 (70/40 wins, $p = 0.014$; +0.022 vs. TS-JEPA, cluster CI $[+0.005, +0.052]$, $p = 1.8 \times 10^{-4}$, median $FPR_{hold} = 0.78\%$), beating system-matched $\max_j z_j$ (0.062, $p = 3 \times 10^{-5}$), raw top-k (0.073, $p = 0.0034$),

TABLE V
CONTEXTUAL VS. POINT ANOMALY DETECTION

(a) Model	NASA Contextual (n=5 seqs) Buf. PR	ER	FPR	f^*=.01	f^*=.05	NASA Point (n=8) f^*=.01	f^*=.05
Laminar-JEPA	0.519	0.775	0.022	0.800	**0.960**	0.783	0.900
Laminar-JEPA-Lite	0.532	**0.875**	0.013	**0.840**	0.920	0.700	0.967
Multi-Horiz. 1D	**0.652**	0.750	**0.006**	0.820	0.920	0.683	0.867
TS-JEPA	0.644	0.675	0.045	0.720	0.840	0.833	0.867
TimesNet	0.317	0.625	0.415	0.560	0.560	0.750	0.767
TranAD	0.162	0.400	0.392	0.400	0.400	0.767	0.800
TCN Obs. Recon.	0.161	0.375	0.387	0.400	0.400	0.800	0.800
ZScoreMax	0.198	0.375	0.558	0.300	0.300	**1.000**	**1.000**

(b) Model	19-Stream Contextual ($\|z\|\leq 5$) q^*=.05	f^*=.01	f^*=.05	19-Stream Point ($\|z\|>5$) q^*=.05	f^*=.01	f^*=.05
Laminar-JEPA	**0.669**	0.448	**0.650**	0.835	0.644	0.795
Laminar-JEPA-Lite	0.655	0.376	0.567	**0.848**	0.581	**0.801**
Multi-Horiz. 1D	0.628	0.498	0.590	0.782	0.683	0.781
TS-JEPA	0.614	**0.520**	0.594	0.776	**0.694**	0.765
TimesNet	0.382	0.290	0.301	0.668	0.549	0.567
TranAD	0.402	0.235	0.300	0.640	0.564	0.611
TCN Obs. Recon.	0.472	0.273	0.297	0.639	0.553	0.563
ZScoreMax	0.376	0.288	0.289	0.673	0.594	0.605

Event recall (ER; autonomous vs. diagnostic f^*); best in **bold**, second underlined. (a) 10 SMAP/MSL streams by NASA taxonomy [4] (contextual: E-1, A-2, A-7, M-1, C-1 [2100, 2210]; point: A-1, P-3, D-3, D-14, D-15, C-1 [550, 750]). (b) n=19 non-GHL streams split into in-range ($\max|z| \leq 5\sigma$) vs. out-of-range ($\max|z| > 5\sigma$) causal events.

TABLE VI
DISJOINT FIVE-BLOCK HELD-OUT ARCHIVE BENCHMARK

(a) Untouched n=118	Pt-F1 ↑ Caus.	Buf.	PR-AUC ↑ Caus.	Buf.	ER ↑ Caus.	Buf.	FPR_{hold} ↓ Caus.	Buf.
TS-JEPA (1D)	.057	.134	.155	.207	.414	.459	**.042**	**.110**
Multi-Horizon JEPA (1D)	.076	.143	**.162**	**.230**	.525	.590	.044	.195
Laminar-JEPA (w/o Quench)	.075	.141	.143	.210	**.644**	.640	.069	.190
Laminar-JEPA (224B)	.072	.144	.143	.211	.609	.619	.068[§]	.190
Laminar-JEPA-Lite (128B)	**.079**[†]	**.156**[‡]	.145	.224	.638[¶]	**.674**	.070	.143
All Archive n=141 (12 **Datasets**)								
TS-JEPA (1D)	.055	.138	.167	.218	.414	.447	**.036**	**.102**
Multi-Horizon JEPA (1D)	.075	.146	**.174**	**.239**	.512	.573	.038	.181
Laminar-JEPA (224B)	.072	.150	.158	.233	.631	.628	.059	.175
Laminar-JEPA-Lite (128B)	**.076**	**.156**	.158	.235	**.636**	**.658**	.060	.124

(b) Held-out	n	Causal Pt-F1 ↑ Lite	Lam.	1D	Buffered Pt-F1 ↑ Lite	Lam.	1D	Causal FPR_{hold} ↓ Lite	Lam.	1D
SMAP	40	.083	.087	**.131**	.186	.180	**.192**	.087	.099	**.083**
SMD	18	.049	**.052**	.031	**.157**	.097	.096	.012	.019	**.003**
Exathlon	17	**.191**	.129	.092	**.204**	.166	.162	.180	.134	**.097**
Daphnet	15	**.072**	.064	.019	**.146**	.146	.101	.068	.077	**.007**
MSL	13	.033	**.055**	.052	.141	.151	**.182**	.012	.017	**.005**
OPPORT.	12	.009	.011	**.046**	.030	**.073**	.045	.021	.008	**.002**

Three seeds under $\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$; best **bold**, second underlined (1D = Multi-Horizon JEPA 1D). [†]Lite vs. 1D (n=118): causal F1 Δ=+0.003 (70/40 wins, p=0.014; vs. TS-JEPA +0.022, cluster CI [+0.005, +0.052], $p=1.8\times10^{-4}$); beats system-matched max-z (.062, $p=3\times10^{-5}$) and raw top-k (.073, p=0.0034). [‡]Buf. F1 Δ=+0.014 (p=0.022; vs. Lam. +0.012, p=0.0064, median buf. FPR_{hold} 4.79% vs. 8.70% 1D). [¶]Causal ER Δ=+0.113 vs. 1D (cluster CI [+0.051, +0.210], 9/9 sets, sign-flip p=0.0039; vs. ℓ_2^2 conf. +0.041, p=0.0041). [§]Disjoint $\mathbf{X}_q$ quench: causal FPR_{hold} $\Delta=-0.0007$ (p=0.010, sign-flip p=0.55), ER −0.035.

and ℓ_2^2 conformal (0.076 F1, +4.1 points ER, p = 0.0041). It lifts buffered Point-F1 from 0.143 to 0.156 (p = 0.022; vs. 0.144 for *Laminar-JEPA*, p = 0.0064, cutting buffered FPR_{hold} from 19.5% to 14.3%, median 8.70% → 4.79%), raises causal ER from 52.5% to 63.8% across all 9/9 datasets (cluster CI [+5.1%, +21.0%], sign-flip p = 0.0039; *Laminar-JEPA* 60.9%, p = 0.0039), and halves causal delay (70.8 vs. 133.5 steps). Without Exathlon (n = 101), *Laminar-JEPA-Lite* holds median causal FPR_{hold} at 0.83% (5.1% mean), leads buffered Point-F1 (0.148 vs. 0.139), and lifts causal ER (50.6% → 58.8%), while Multi-Horizon 1D leads short-stream causal Point-F1 (0.073 vs. 0.060). Disjoint $\mathbf{X}_q$ quenching trims 4H causal FPR_{hold} by only 0.07 points (6.90% → 6.83%, p = 0.010, sign-flip p = 0.55) at −3.5 points ER (unweighted N_{224} shifts PR-AUC by ≤ 0.0016), reinforcing *Laminar-JEPA-Lite*'s zero-quench design.

2) *Autocorrelation, sample size, and cross-fitting:* Across the 19 development streams, median Sokal τ_{int} (IQR) is 2.51 ([1.52, 4.04], max 63.5) for z_j, 1.88 ([1.22, 2.73]) for N_{edge}, 5.58 ([3.07, 14.09], max 206.0) for N_{224}, and 6.75 ([5.10, 16.88]) for 1D scores. At median N_{cal} = 408, median N_{eff} (expected tail count) is 154.6 (2.32) at $q_{0.985}$, 67.1 (0.34) at $q_{0.995}$, and 195.9 (0.20) at $q_{0.999}$; even N_{eff} = 300 yields only 4.5–0.3 tail points. Block-bootstrap ($L_{\text{blk}} \in \{1..128\}$) 95% CI half-widths span 12.7%–25.0%, while floors (5.0, 2.5) hold median FPR_{hold} to 0.36% and 0.00%. On n = 118, N_{cal} < 250 (n = 12) averages 16.0% causal FPR_{hold} for *Laminar-JEPA-Lite* (17.6% *Laminar-JEPA*) versus 1.05% (1.60%) for $N_{\text{cal}} \geq$ 2,500 (n = 22); on short histories (T_{nom} < 2,500, n = 18), two-fold cross-fitting cuts causal FPR_{test} from 15.4% to 3.8% at 69.9% ER, while Beta quantiles and IQR extrapolation remain exploratory.

3) *Regime drift vs. tail misfit via permutation decomposition:* Refitting thresholds across chronological (FPR_{chr}), L = 64 block-permuted (FPR_{blk}), and pointwise i.i.d. ($\text{FPR}_{\text{pt}} \leq$ 0.35%) splits is consistent with inter-block regime drift without ruling out parametric tail misfit: $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$ represents 71.3% of excess causal FPR in TS-JEPA and 78.3%–81.8% in reconstructors, versus 18.7% in the centered model. Within-checkpoint traces of *Laminar-JEPA* (n=6, $K \in \{9, 25, 55\}$) show pre-tap vectors clustering on an off-center cap ($\|\bar{\boldsymbol{\mu}}_{\text{ctx}}\|_2 \in [4.73, 5.18]$ vs. $\sqrt{32} \approx 5.66$) at 0.06–0.16 anomaly divergence, whereas tap fusion and barycenter subtraction restore divergence to 0.70–0.94 and cut mean coordinate D_{KS} from 0.140 to 0.096.

VIII. CONCLUSION AND FUTURE WORK

We evaluated a two-tier family of multi-horizon circuit-breaker architectures (*Laminar-JEPA-Lite*, preferred deployment tier; *Laminar-JEPA*, 224-dial reference) for abrupt and intermediate-onset telemetry faults across 33 development and 118 untouched held-out streams:

1) *Preferred distilled vs. reference panels:* On the 33-stream benchmark, *Laminar-JEPA* and *Laminar-JEPA-Lite* reach 0.244 and 0.242 buffered Point-F1 and 67.3% and 65.0% causal ER at 1.3% and 1.0% FPR_{hold}, while *Laminar-JEPA-Lite* cuts parameters by 35%–52%, accelerates inference by 2.43×, and leads 7-dataset macro Point-F1 (0.171; 19-stream cache 0.071 and 0.098 vs. 0.013 1D). Across 118 untouched streams ($\mathbf{X}_{\text{sel}} \perp \mathbf{X}_q \perp \mathbf{X}_{\text{cal}}$), *Laminar-JEPA-Lite* lifts causal Point-F1 (0.076 → 0.079, p = 0.014), buffered Point-F1 (0.143 → 0.156, p = 0.022), and causal ER (52.5% → 63.8%, 9/9 sets, p = 0.0039), beating system-matched $\max_j z_j$ (0.062) and raw top-k (0.073), while disjoint quenching trims 4H causal FPR_{hold} by only 0.07 points (p = 0.55 sign-flip).
2) *Mechanisms and latency:* Matched-backbone controls confirm LayerNorm tail compression and barycenter drift reduction while scalar R_{LT} modulation is null, and five-stream sweeps ($S \in \{1..64\}$) retain 2.1 × –4.3× causal Point-F1 gains down to S = 1.
3) *Calibration, single-sensor, and ramp limits:* Sokal autocorrelation ($\tau_{\text{int}} \leq$ 206) leaves 1.5 and 0.3 expected tail points at N_{eff} = 300, though two-fold cross-fitting cuts short-stream (T_{nom} < 2,500) causal FPR_{test} from 15.4%

to 3.8%. An optional channel branch lifts injected single-sensor ER (80.7%–96.5%) only at 6.8%–19.4% FPR_{test}, and excess causal detection vanishes on slow ramps once $L \geq 1024$ ($4C$); pairing raw-level trend charts and testing natural sensor failures remain future priorities.

REFERENCES

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] M. P. Cuéllar, M. Santos, J. L. Alonso, M. Fabregas, and R. Farías, "Explainable anomaly detection in spacecraft telemetry," *Engineering Applications of Artificial Intelligence*, vol. 133, p. 108119, 2024.

[4] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[8] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[9] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[10] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.

[11] J. L. Ba, J. R. Kiros, and G. E. Hinton, "Layer normalization," *arXiv preprint arXiv:1607.06450*, 2016.

[12] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.

[13] K. Zhang, Q. Wen, C. Zhang, R. Cai, M. Wang, L. Liu, and L. Sun, "Self-supervised time series representation learning: A systematic survey," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 46, no. 5, pp. 3241–3261, 2024.

[14] E. Eldele, M. Ragab, Z. Chen, M. Wu, C. K. Kwoh, X. Li, and C. Guan, "Time-series representation learning via temporal and contextual contrasting," in *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2021, pp. 2352–2359.

[15] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.

[16] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.

[17] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.

[18] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.

[19] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.

[20] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.

[21] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.

[22] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.

[23] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.

[24] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.

[25] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.

[26] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.

[27] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.

[28] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.

[29] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.

[30] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.

[31] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.

[32] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.

[33] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.

[34] O. Reynolds, "On the dynamical theory of incompressible viscous fluids and the determination of the criterion," *Philosophical Transactions of the Royal Society of London A*, vol. 186, pp. 123–164, 1895.

[35] H. Hotelling, "Multivariate quality control, illustrated by the air testing of sample bombsights," in *Techniques of Statistical Analysis*. McGraw-Hill, 1947, pp. 111–184.

[36] C. A. Lowry, W. H. Woodall, C. W. Champ, and S. E. Rigdon, "A multivariate exponentially weighted moving average control chart," *Technometrics*, vol. 34, no. 1, pp. 46–53, 1992.

[37] A. A. Balkema and L. de Haan, "Residual life time at great age," *The Annals of Probability*, vol. 2, no. 5, pp. 792–804, 1974.

[38] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.

[39] E. S. Page, "Continuous inspection schemes," *Biometrika*, vol. 41, no. 1/2, pp. 100–115, 1954.

[40] Y. Benjamini and Y. Hochberg, "Controlling the false discovery rate: a practical and powerful approach to multiple testing," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 57, no. 1, pp. 289–300, 1995.

[41] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.

[42] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.

[43] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.

[44] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.

[45] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.

[46] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.

[47] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.

[48] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," *arXiv preprint arXiv:1803.01271*, 2018.

[49] Q. Garrido, R. Balestriero, L. Najman, and Y. LeCun, "RankMe: Assessing the downstream performance of pretrained self-supervised representations by their rank," in *International Conference on Machine Learning (ICML)*, 2023, pp. 10 929–10 974.

[50] E. H. Lieb and W. E. Thirring, "Inequalities for the moments of the eigenvalues of the Schrödinger Hamiltonian and their relation to Sobolev inequalities," in *Studies in Mathematical Physics: Essays in Honor of Valentine Bargmann*. Princeton University Press, 1976, pp. 269–303.

[51] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.

[52] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *International Conference on Learning Representations (ICLR)*, 2019.